

CSCI465/ECEN433: Introduction to Parallel Computing

Research Project (Automatic Parallelization)

Introduction

Automatic parallelization is a technique used in parallel computation to transform sequential code into parallel code automatically, without programmer intervention. It involves analyzing the code to identify opportunities for parallelism, dividing the computation into parallel tasks, and coordinating their execution to achieve performance increases. Automatic parallelization is a complex procedure that necessitates sophisticated algorithms and tools to guarantee that the parallelized code is correct and efficient. As the demand for high-performance computation increases and parallel architectures become more prevalent in modern computing systems, its significance has grown. Automatic parallelization can be used to maximize the potential of parallel architectures, reduce computation time, and enhance application scalability. However, it presents several challenges, such as the need to balance workload distribution and reduce communication overhead, making it a subject of ongoing research in the field of parallel computing.

Accordingly, the objective of this project is to introduce undergraduate students to the concept of automatic parallelization in parallel computing. The project will require students to **implement an auto parallelization technique** in a given codebase, measure the performance gains achieved, and report their findings.

- To help you to start, you can read the references available through this [link](#).

Requirements:

- Understanding of parallel computing and programming concepts.
- Knowledge of programming languages such as C/C++, Python, or Java.
- Familiarity with tools like OpenMP, MPI, and ISPC.
- Access to a computing environment with multiple cores or processors.

Objectives:

- Understand the codebase for the auto parallelization exercise. The codebase should be a non-parallel code written in a programming language such as C/C++, Python, or Java.
- Identify opportunities for parallelization in the codebase. This can be achieved by analyzing the code and profiling it using tools such as Gprof, Perf, and VTune.
- Implement auto parallelization techniques using tools such as OpenMP, MPI, or ISPC.
- Measure the performance gains achieved using tools such as Gprof, Perf, and VTune. Compare the performance of the original code with the parallelized code and report their findings.
- Document their findings in a report that includes a description of the codebase, the identified opportunities for parallelization, the implemented auto parallelization techniques, the performance gains achieved, and a discussion of the results.

Deliverables

The main deliverable in this project is a **term paper**. You are expected to write a **4–6 page** (in 10pt IEEE style) paper on any topic relevant to Automatic parallelization. **You can work in groups of 4-6 students on the project.**

- Term Paper - 100%
 - Phase I: Final Term Paper – 15 Marks
 - Phase II: Presentation and Discussion – 5 Marks

➤ Phase I: Final Term Paper (15 Marks)

Each group is expected to submit a 4-6 page paper describing their progress in 10pt IEEE format (PDF). The paper should clearly address the following questions:

1. Title and Abstract (summarize the paper's main insights and be catchy)
2. Introduction (e.g., motivation for the problem, problem definition, main contributions)
3. Related Work
4. Proposed Solution
5. Evaluation and Results
6. Conclusion and Future Work

➤ Phase II: Presentation and Discussion (5 Marks)

1. Time management
2. Presentation Clarity
3. Q and A.

Tools and links

- Search for Published Research
 - [EKB](#)
 - [IEEE Xplore Digital Library](#)
 - [ACM Digital Library](#)
 - [ScienceDirect](#)
 - [SpringerLink](#)
 - [Google Scholar](#)
- Figures and flowcharts
 - [Gliffy](#)
 - [Draw.io](#)
- Writing
 - [Mendeley](#)
 - [Overleaf](#)
 - [EndNote](#)
 - [Ms Word](#)
- Spelling and grammar
 - [Grammarly](#)

Important Dates

- **Sunday, May 17, 2026** - Submission of research proposal
- **TBA** - Presentation and Discussion

Best wishes